

M3. Generative Models with Other Statistical Distance

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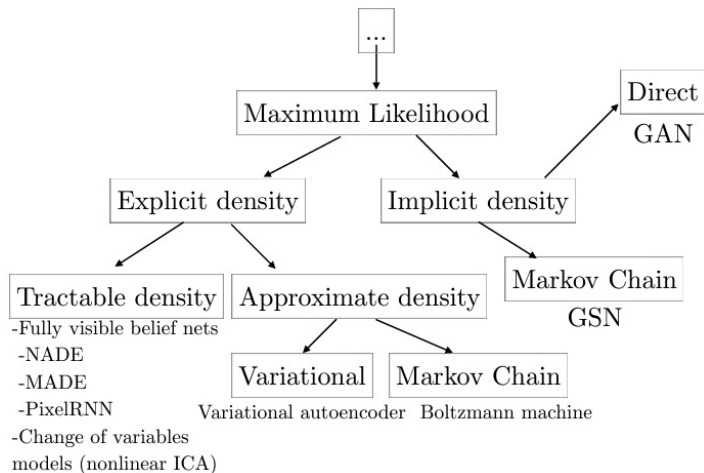
Department of Mathematical Sciences

MTH345: Deep Generative Models: A Statistical Perspective



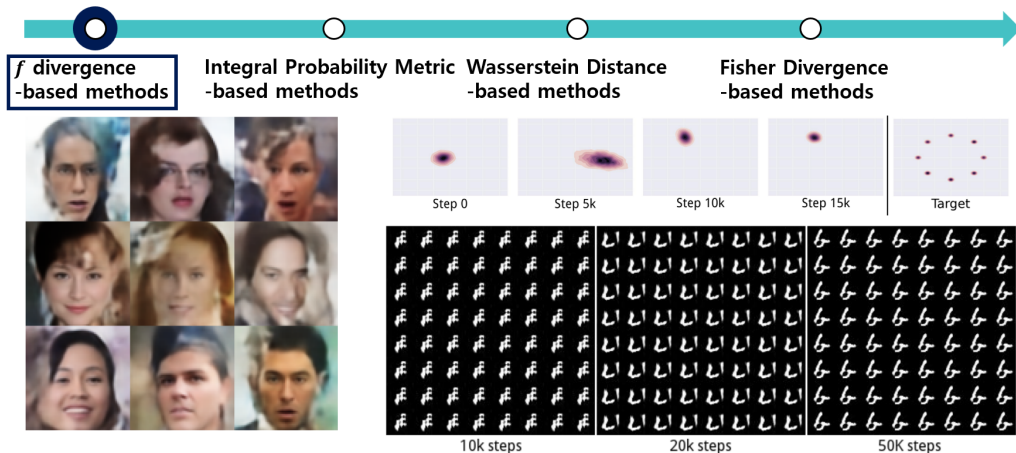
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SCIENCE AND TECHNOLOGY

Introduction



The figure is from Goodfellow (2016).

Introduction



(a) Blurry generation results from VAEs

(b) Mode collapse results from GANs

Images are edited from Tolstikhin et al. (2018) and Metz et al. (2017).

Introduction



- ▶ The mode collapse phenomenon in GANs demonstrates that p_θ fails to capture the support of p .
- ▶ Several works have criticized the f -divergence,

$$\mathcal{D}_f(p\|p_\theta) = \int f\left(\frac{p(\mathbf{x})}{p_\theta(\mathbf{x})}\right) p_\theta(\mathbf{x}) d\mathbf{x}, \quad (1)$$

pointing out that the density ratio, $p(\mathbf{x})/p_\theta(\mathbf{x})$, is not well-defined when p places mass on a region where p_θ has (almost) zero density.

Lecture Topics

- ▶ In this module, we study several alternative statistical distances in the deep generative model literature.
- ▶ Recommended reference materials include:
 - ▶ Chap. 9: Gretton et al. (2012), “A kernel two-sample test,” *The Journal of Machine Learning Research*.
 - ▶ Chap. 10: Santambrogio (2015), “Optimal Transport for Applied Mathematicians: Calculus of Variations, PDEs, and Modeling,” *Birkhäuser*.
 - ▶ Chap. 11: Hyvärinen (2005), “Estimation of non-normalized statistical models by score matching,” *The Journal of Machine Learning Research*.

Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

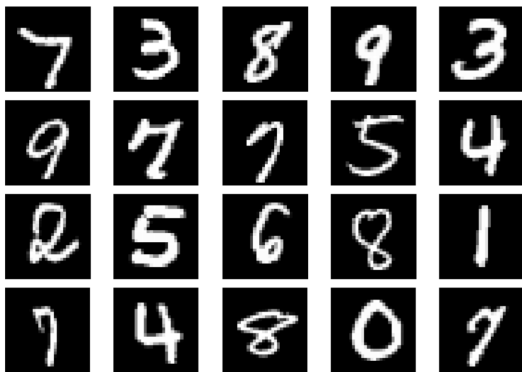
5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

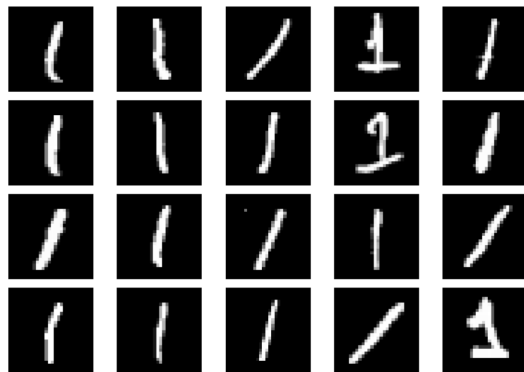
9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Motivation

(i) Group 1: All Digits



(ii) Group 2: Digit '1'



- Q: How to quantify the difference between two data collections?

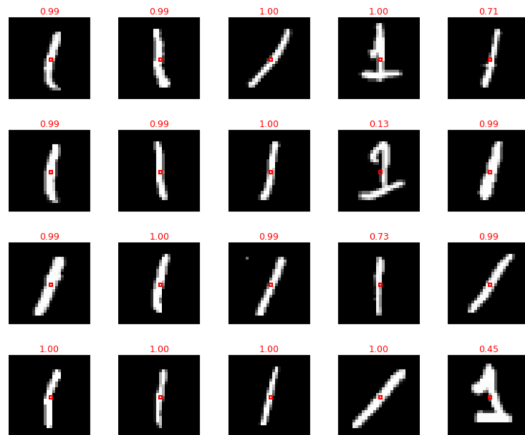
Motivation

(i) Group 1: All Digits



Group 1 Mean = 0.3906

(ii) Group 2: Digit '1'



Group 2 Mean = 0.8965

Integral Probability Metric



- ▶ The Integral Probability Metric (IPM) between $p(\mathbf{x})$ and $q(\mathbf{x})$ is defined as:

$$\gamma_{\mathcal{F}}(p, q) := \sup_{f \in \mathcal{F}} \left| \int_{\mathcal{X}} f(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} - \int_{\mathcal{X}} f(\mathbf{x}) q(\mathbf{x}) d\mathbf{x} \right|, \quad (2)$$

where $\mathcal{F}$ is a class of real-valued functions.

- ▶ For $\gamma_{\mathcal{F}}(p, q)$ to be a valid metric, the function class $\mathcal{F}$ must be sufficiently rich (e.g., the set of all bounded continuous functions or a universal RKHS unit ball).

Integral Probability Metric

- ▶ **Total Variation Distance:** The total variation distance, $\delta(p, q) := \frac{1}{2} \int_{\mathcal{X}} |p(\mathbf{x}) - q(\mathbf{x})| d\mathbf{x}$, has an alternative expression as an IPM:

$$\delta(p, q) = \sup_{f \in \mathcal{F}_{\text{ind}}} \left| \int_{\mathcal{X}} f(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} - \int_{\mathcal{X}} f(\mathbf{x}) q(\mathbf{x}) d\mathbf{x} \right|, \quad (3)$$

where $\mathcal{F}_{\text{ind}} = \{f : f(\mathbf{x}) = I(\mathbf{x} \in A) \text{ for some } A \in \mathcal{A}\}$ is the class of indicator functions for all (measurable) events.

- ▶ **Earth Mover's Distance:** When $\mathcal{F}$ consists of all 1-Lipschitz continuous functions, the corresponding IPM becomes the Earth mover's distance (or 1-Wasserstein distance). Details will be discussed in the subsequent chapter.

Integral Probability Metric

- ▶ We denote the kernel function by $k(\mathbf{x}, \mathbf{x}')$. For any positive definite kernel, Mercer's theorem guarantees the existence of a kernel feature map φ such that

$$k(\mathbf{x}, \mathbf{x}') = \langle \varphi(\mathbf{x}), \varphi(\mathbf{x}') \rangle. \quad (4)$$

- ▶ For example, when $d = 1$, the quadratic kernel $k_{\text{quad}}(x, x') = (xx' + c)^2$ can be written as the inner product:

$$k_{\text{quad}}(x, x') = \varphi(x)^\top \varphi(x'), \quad (5)$$

where $\varphi(x) = (c, \sqrt{2cx}, x^2)^\top$.

- ▶ As another example, for the radial basis function (RBF) kernel $k_{\text{RBF}}(x, x') = \exp\left(-\frac{(x-x')^2}{2\sigma^2}\right)$,

$$\varphi(x) = \exp\left(-\frac{x^2}{2\sigma^2}\right) \left(1, \frac{x}{\sigma\sqrt{1!}}, \frac{x^2}{\sigma^2\sqrt{2!}}, \dots\right)^\top. \quad (6)$$

Integral Probability Metric

- ▶ Maximum Mean Discrepancy (MMD): The MMD between $p(\mathbf{x})$ and $q(\mathbf{x})$ can be expressed as¹

$$\text{MMD}_k(p, q) = \left\| \int_{\mathcal{X}} \varphi(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} - \int_{\mathcal{X}} \varphi(\mathbf{x}) q(\mathbf{x}) d\mathbf{x} \right\|_{\mathcal{H}} \quad (7)$$

where $\mathcal{H}$ denotes the reproducing kernel Hilbert space (RKHS) and $\|\cdot\|_{\mathcal{H}}$ denotes the norm in $\mathcal{H}$.

- ▶ By the reproducing property, the squared MMD can be expanded as:

$$\begin{aligned} \text{MMD}_k^2(p, q) = & \int_{\mathcal{X} \times \mathcal{X}} k(\mathbf{x}, \mathbf{x}') p(\mathbf{x}) p(\mathbf{x}') d\mathbf{x} d\mathbf{x}' - 2 \int_{\mathcal{X} \times \mathcal{X}} k(\mathbf{x}, \mathbf{y}) p(\mathbf{x}) q(\mathbf{y}) d\mathbf{x} d\mathbf{y} \\ & + \int_{\mathcal{X} \times \mathcal{X}} k(\mathbf{y}, \mathbf{y}') q(\mathbf{y}) q(\mathbf{y}') d\mathbf{y} d\mathbf{y}'. \end{aligned} \quad (8)$$

¹The MMD is defined as the distance between kernel means in $\mathcal{H}$.

Generative Moment Matching Network

- ▶ Generative Moment Matching Networks (GMMNs) (Li et al., 2015) use an empirical MMD as a loss, avoiding the need for an adversarial network as in GANs.
- ▶ Given minibatch samples $(\mathbf{x}_i)_{i=1}^B$ and $(G_\theta(\mathbf{z}_i))_{i=1}^B$, the loss can be expressed as:

$$\begin{aligned}\widehat{\text{MMD}}_k^2(p(\mathbf{x}), p_\theta(\mathbf{x})) = & \frac{1}{B(B-1)} \sum_{i=1}^B \sum_{j \neq i}^B k(\mathbf{x}_i, \mathbf{x}_j) - \frac{2}{B^2} \sum_{i=1}^B \sum_{j=1}^B k(\mathbf{x}_i, G_\theta(\mathbf{z}_j)) \\ & + \frac{1}{B(B-1)} \sum_{i=1}^B \sum_{j \neq i}^B k(G_\theta(\mathbf{z}_i), G_\theta(\mathbf{z}_j)).\end{aligned}\tag{9}$$

- ▶ GMMNs used a mixture of multiple Gaussian kernels with various bandwidth parameters.

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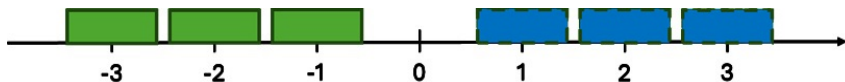
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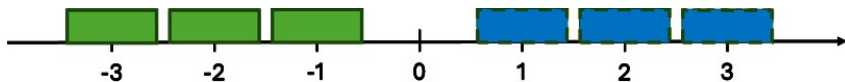
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Motivation: Monge Problem



- ▶ *Optimal Transport* is a problem focused on identifying maps that most efficiently move one distribution of mass to another and investigating their properties.
- ▶ Let's consider a toy example of moving green bold boxes on the left to blue dashed-box regions on the right.
- ▶ Which way is more efficient?
 1. T_1 : $T_1(-3) = 1$, $T_1(-2) = 2$, and $T_1(-1) = 3$
 2. T_2 : $T_2(-3) = 3$, $T_2(-2) = 2$, and $T_2(-1) = 1$

Motivation: Monge Problem



- The answer depends on criteria. When we consider the squared L_2 -distance between the original green boxes and their transportation results:

$$\begin{aligned} & \left((-3 - T_1(-3))^2 + (-2 - T_1(-2))^2 + (-1 - T_1(-1))^2 \right) / 3 \\ &= \left((-3 - 1)^2 + (-2 - 2)^2 + (-1 - 3)^2 \right) / 3 = 16 \end{aligned} \tag{10}$$

$$\begin{aligned} & \left((-3 - T_2(-3))^2 + (-2 - T_2(-2))^2 + (-1 - T_2(-1))^2 \right) / 3 \\ &= \left((-3 - 3)^2 + (-2 - 2)^2 + (-1 - 1)^2 \right) / 3 = 56/3 \end{aligned} \tag{11}$$

Q: How many maps are there? What happens when we consider the L_1 -distance?

Motivation: Monge Problem

- Monge Problem: Given $p(\mathbf{x})$ and $q(\mathbf{x})$ and a cost function $c : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, the optimal transportation cost is defined as:²

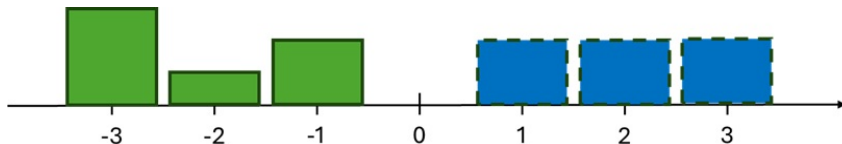
$$\inf_{T: T_{\#}p=q} \int_{\mathcal{X}} c(\mathbf{x}, T(\mathbf{x})) p(\mathbf{x}) d\mathbf{x}, \quad (12)$$

where the constraint $T_{\#}p = q$ represents the push-forward operation, i.e., $p(\mathbf{x}) = q(T(\mathbf{x})) |\det J_T(\mathbf{x})|$. An optimal solution T^* to this problem, if it exists, is referred to as an *optimal transport map*.

- When $c(\mathbf{x}, \mathbf{x}') = \|\mathbf{x} - \mathbf{x}'\|^2$, T^* exists and is unique. In the univariate case, T^* maps percentiles of $p(\mathbf{x})$ to the corresponding percentiles of $q(\mathbf{x})$.

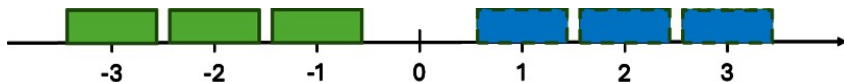
²Optimal transport is more generally formulated with probability measures, but for simplicity, we state definitions using PDFs in this course.

Motivation: Kantorovich Problem



- ▶ The constraint on the optimal transport map, $T_{\#}p = q$, generally poses challenges in defining the optimal transportation cost.
- ▶ The *Kantorovich Problem* is a generalization of the Monge Problem to alleviate these challenges.

Motivation: Kantorovich Problem

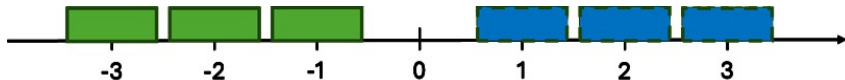


- In the first example, the joint distribution of $(x, T_1(x))$ can be expressed as:

(Source, Target)	1	2	3	Total
-3	1/3	0	0	1/3
-2	0	1/3	0	1/3
-1	0	0	1/3	1/3
Total	1/3	1/3	1/3	1

- That is, each (deterministic) transport map identifies the corresponding joint distribution of the source data and their transported results.

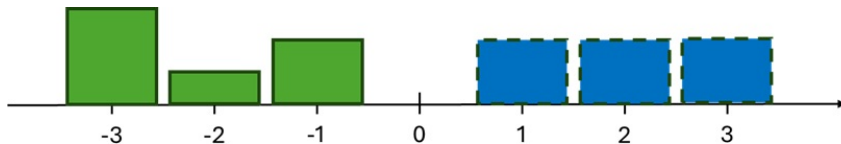
Motivation: Kantorovich Problem



- ▶ The Kantorovich Problem involves minimizing transportation costs among all joint distributions whose marginals are the source and target data distributions:

(Source, Target)	1	2	3	Total
-3	$p_{(-3,1)}$	$p_{(-3,2)}$	$p_{(-3,3)}$	$1/3$
-2	$p_{(-2,1)}$	$p_{(-2,2)}$	$p_{(-2,3)}$	$1/3$
-1	$p_{(-1,1)}$	$p_{(-1,2)}$	$p_{(-1,3)}$	$1/3$
Total	$1/3$	$1/3$	$1/3$	1

Motivation: Kantorovich Problem



(Source, Target)	1	2	3	Total
-3	$p(-3,1)$	$p(-3,2)$	$p(-3,3)$	$1/2$
-2	$p(-2,1)$	$p(-2,2)$	$p(-2,3)$	$1/6$
-1	$p(-1,1)$	$p(-1,2)$	$p(-1,3)$	$1/3$
Total	$1/3$	$1/3$	$1/3$	1

Motivation: Kantorovich Problem

$$\min_{p_{(-3,1)}, \dots, p_{(-1,3)}} \left(c(-3, 1)p_{(-3,1)} + \dots + c(-1, 3)p_{(-1,3)} \right)$$

subject to

$$p_{(-3,1)} + p_{(-3,2)} + p_{(-3,3)} = 1/2,$$

$$p_{(-2,1)} + p_{(-2,2)} + p_{(-2,3)} = 1/6,$$

$$p_{(-1,1)} + p_{(-1,2)} + p_{(-1,3)} = 1/3,$$

$$p_{(-3,1)} + p_{(-2,1)} + p_{(-1,1)} = 1/3,$$

$$p_{(-3,2)} + p_{(-2,2)} + p_{(-1,2)} = 1/3,$$

$$p_{(-3,3)} + p_{(-2,3)} + p_{(-1,3)} = 1/3,$$

$$p_{(-3,1)} \geq 0, \dots, p_{(-1,3)} \geq 0.$$

(13)

Application to High-dimensional Data

			
	1/3	0	0
	0	1/3	0
	0	0	1/3

Joint Distribution

			
	0.10	0.30	0.40
	0.40	0.05	0.25
	0.25	0.30	0.10

Transportation Cost

$$\begin{aligned}
 & (0.10 \times (1/3) + 0.30 \times 0 + 0.40 \times 0) \\
 & + (0.40 \times 0 + 0.05 \times (1/3) + 0.25 \times 0) \\
 & + (0.25 \times 0 + 0.30 \times 0 + 0.10 \times (1/3)) = 0.08
 \end{aligned}$$

Wasserstein Distance



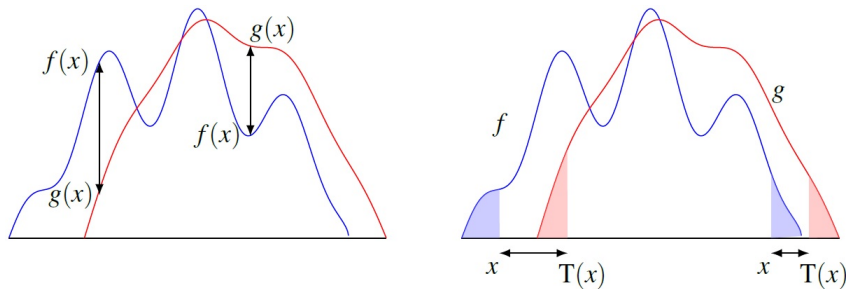
- The p -Wasserstein distance is defined as:

$$W_p(p, q) := \left(\inf_{\pi \in \Pi(p, q)} \int_{\mathcal{X} \times \mathcal{X}} d^p(\mathbf{x}, \mathbf{x}') d\pi(\mathbf{x}, \mathbf{x}') \right)^{1/p},$$

where $p \in [1, \infty)$, d is a metric defined on $\mathcal{X}$,³ and $\Pi(p, q)$ denotes the set of all joint distributions (couplings) whose marginal densities are p and q .

³The Kantorovich problem is generally formulated using a general cost function $c(\mathbf{x}, \mathbf{x}')$.

Wasserstein Distance



- ▶ The f -divergence utilizes density-ratios to quantify discrepancies between distributions.
- ▶ In contrast, Wasserstein distances utilize transportation costs.

The figure is from Santambrogio (2015).

Wasserstein Distance

- ▶ When $p = 1$ (often called the *Earth Mover's Distance*), the Kantorovich-Rubinstein duality holds (Villani et al., 2009; Villani, 2021):

$$W_1(p, q) = \sup_{\substack{f, g: \\ f(\mathbf{x}) - g(\mathbf{x}') \leq d(\mathbf{x}, \mathbf{x}')}} \left(\int_{\mathcal{X}} f(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} - \int_{\mathcal{X}} g(\mathbf{x}') q(\mathbf{x}') d\mathbf{x}' \right). \quad (14)$$

- ▶ When $d(\mathbf{x}, \mathbf{x}') = \|\mathbf{x} - \mathbf{x}'\|$ for some norm $\|\cdot\|$, the dual form simplifies to:

$$W_1(p, q) = \sup_{\text{Lip}(f) \leq 1} \left(\int_{\mathcal{X}} f(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} - \int_{\mathcal{X}} f(\mathbf{x}) q(\mathbf{x}) d\mathbf{x} \right), \quad (15)$$

where $\text{Lip}(f) := \sup_{\mathbf{x} \neq \mathbf{x}'} \frac{|f(\mathbf{x}) - f(\mathbf{x}')|}{\|\mathbf{x} - \mathbf{x}'\|}$ denotes the Lipschitz constant of f . This formulation confirms that W_1 is an Integral Probability Metric (IPM).

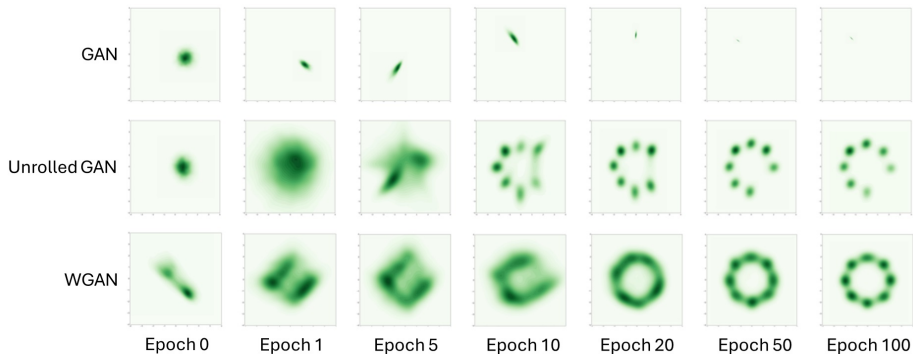
Wasserstein Generative Adversarial Networks

- ▶ Wasserstein Generative Adversarial Networks (WGANs) (Arjovsky et al., 2017) parameterize 1-Lipschitz continuous functions using a neural network f_ϕ , optimizing:

$$\min_{\theta} \max_{\phi} \left(\int_{\mathcal{X}} f_{\phi}(\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} - \int_{\mathcal{X}} f_{\phi}(\mathbf{x}) p_{\theta}(\mathbf{x}) d\mathbf{x} \right). \quad (16)$$

- ▶ If the set $\{f_{\phi} \mid \phi \in \Phi\}$ perfectly captures all 1-Lipschitz functions $\{f \mid \text{Lip}(f) \leq 1\}$, Equation (16) equals $\min_{\theta} W_1(p_{\text{data}}, p_{\theta})$.
- ▶ If the condition is relaxed to C -Lipschitz continuity, Equation (16) equals $C \cdot \min_{\theta} W_1(p_{\text{data}}, p_{\theta})$.
- ▶ To enforce the Lipschitz constraint during training, WGANs clip the network weights to a compact interval $[-c, c]$.

Wasserstein Generative Adversarial Networks



- ▶ WGANs performed better than GAN baselines in learning the distribution of Gaussian mixtures with 8 modes.

Images are edited from Arjovsky et al. (2017).

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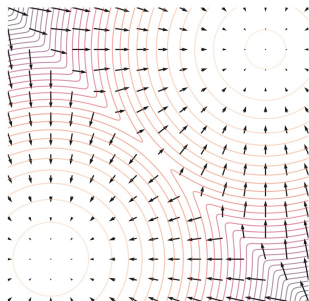
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Fisher Divergence



- ▶ IPM and Wasserstein distance-based methods have alleviated optimization issues; however, adversarial training is still practically difficult.
- ▶ Recent works have focused on score functions instead of densities, using estimated scores to generate data.

Fisher Divergence

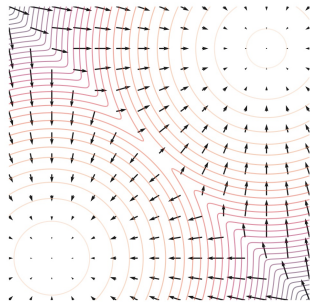


- Fisher divergence (Johnson, 2004) is the expected difference between the (Stein) scores (Liu et al., 2016) of two distributions. It can be expressed as:

$$\text{FD}(p \parallel p_\theta) = \int \|\nabla_{\mathbf{x}} \log p(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_\theta(\mathbf{x})\|^2 p(\mathbf{x}) d\mathbf{x}. \quad (17)$$

Visualization of $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ of a mixture of two Gaussians, from Song (2021).

Fisher Divergence



- It is zero if and only if $p = q$.

Proof:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \nabla_{\mathbf{x}} \log q(\mathbf{x}) \implies p(\mathbf{x}) = Cq(\mathbf{x}) \quad (18)$$

and $C = 1$ because $\int p(\mathbf{x}) d\mathbf{x} = \int q(\mathbf{x}) d\mathbf{x} = 1$.

Visualization of $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ of a mixture of two Gaussians, from Song (2021).

Fisher Divergence

- ▶ The Fisher divergence is useful for learning energy-based models (Teh et al., 2003), such as Boltzmann distributions:

$$p_{\theta}(\mathbf{x}) = \frac{1}{C(\theta)} \exp(-E_{\theta}(\mathbf{x})) \quad (19)$$

where $C(\theta) := \int \exp(-E_{\theta}(\mathbf{x})) d\mathbf{x}$. In this case,

$$\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) = -\nabla_{\mathbf{x}} E_{\theta}(\mathbf{x}) \quad (20)$$

holds, and the normalizing constant disappears.

- ▶ Under certain conditions, the minimizer of the Fisher-divergence is consistent. See Hyvärinen (2005) for further details on asymptotic properties of score-based estimators.

Data Generation using Score Function

- ▶ When we model $p_{\theta}(\mathbf{x})$ as a parametric family distribution and use the minimizer of $\text{FD}(p||p_{\theta})$ as an estimator, we can directly generate data from $p_{\theta}(\mathbf{x})$ with the estimator.
- ▶ How to generate data using score functions $S(\mathbf{x})$ in general?
- ▶ The key idea is to introduce *Langevin dynamics* in the sampling process.

Data Generation using Score Function

- Langevin dynamics describes the stochastic movement of a fluid particle located at $\mathbf{X}(t)$:

$$m \frac{d^2 \mathbf{X}(t)}{dt^2} = -\nabla_{\mathbf{x}=\mathbf{X}(t)} U(\mathbf{x}) - \lambda \frac{d\mathbf{X}(t)}{dt} + \sqrt{2\lambda k_B T} \mathbf{B}(t), \quad (21)$$

where m is the mass, U is the potential functions, λ is the damping coefficient, k_B is the Boltzmann constant, T is the temperature, and $\mathbf{B}(t)$ represents the Brownian motion.

- In the overdamped case, where the inertial force is negligible, when $\lambda = 1$, we get

$$d\mathbf{X}(t) = -\nabla_{\mathbf{x}=\mathbf{X}(t)} U(\mathbf{x}) dt + \sqrt{2k_B T} d\mathbf{B}(t) \quad (22)$$

where $d\mathbf{B}(t) \sim N(0, dtI_m)$.⁴ Its stationary distribution is the Boltzmann distribution with energy $U/(k_B T)$, $p(\mathbf{x}(\infty)) \propto \exp(-U(\mathbf{x}(\infty))/(k_B T))$.

⁴This is a special case of the Itô drift-diffusion process.

Data Generation using Score Function

- By substituting $U(\mathbf{x}) = -\log p(\mathbf{x})$ and setting $T = 1/k_B$, we obtain:

$$d\mathbf{X}(t) = \nabla_{\mathbf{x}} \log p(\mathbf{x}) dt + \sqrt{2dt} \mathcal{E}(t), \quad (23)$$

where $\mathcal{E}(t) \sim N(0, I_m)$, and the corresponding stationary distribution is $p(\mathbf{x})$.

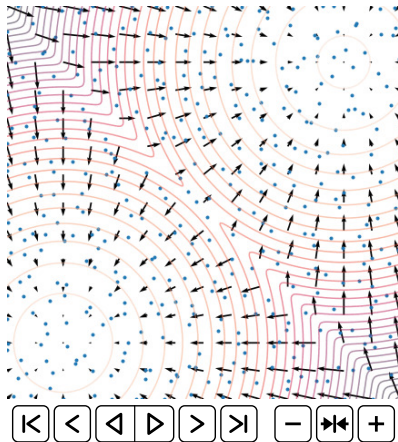
- The discrete approximation with $dt = \eta/2$ and $S_{\theta^*}(\mathbf{x})$ results in the following iterative sampling process:

$$\mathbf{X}(t) = \mathbf{X}(t-1) + (\eta/2)S_{\theta^*}(\mathbf{X}(t-1)) + \sqrt{\eta}\mathcal{E}(t), \quad (24)$$

where $\mathbf{X}(T)$ approximately follows $p(\mathbf{x})$.

- See the following materials for further details:
 1. A lecture by Dr. Kirill Neklyudov
<https://www.youtube.com/watch?v=3-KzIjoFJy4>
 2. Section 'Recovering Boltzmann Statistics' at
https://en.wikipedia.org/wiki/Langevin_equation

Data Generation using Score Function



The video is from Song (2021).

Sliced Score Matching

- ▶ Score matching estimation (Hyvärinen, 2005) was proposed targeting Fisher divergence in learning distributions.
- ▶ Let $S_\theta(\mathbf{x}) := \nabla_{\mathbf{x}} \log p_\theta(\mathbf{x})$. Then,

$$\text{FD}(p||p_\theta) = \int (\text{tr}(\nabla_{\mathbf{x}} S_\theta(\mathbf{x})) + \frac{1}{2} \|S_\theta(\mathbf{x})\|^2) p(\mathbf{x}) d\mathbf{x} \quad (25)$$

up to a constant addition and sign-preserving multiplication. We assume that $S_\theta(\mathbf{x})p(\mathbf{x})$ vanishes at the boundary, e.g., $(x_1, \dots, x_{i-1}, \pm\infty, x_{i+1}, \dots, x_m)$.

- ▶ In the objective, the Hessian term $\nabla_{\mathbf{x}} S_\theta(\mathbf{x})$ poses another computational challenge.

Sliced Score Matching

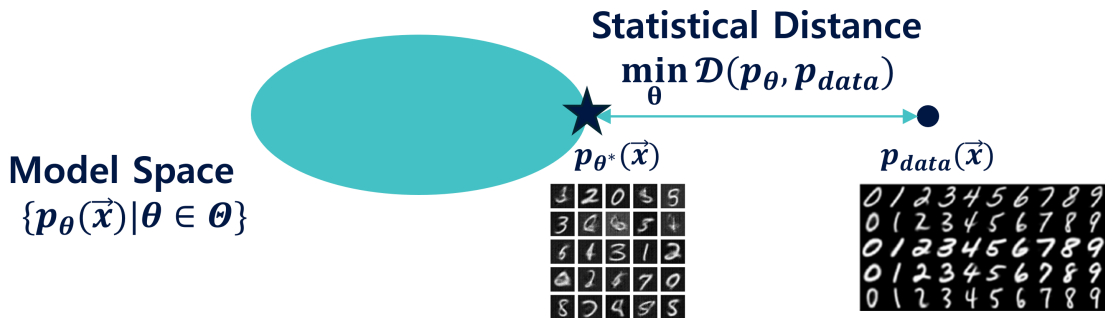
- ▶ Sliced score matching (Song et al., 2020) targets sliced Fisher divergence (SFD) to overcome this limitation:

$$\text{SFD}(p||p_\theta) = \int (\mathbf{v}^T \nabla_{\mathbf{x}} S_\theta(\mathbf{x}) \mathbf{v} + \frac{1}{2} (\mathbf{v}^T S_\theta(\mathbf{x}))^2) p(\mathbf{x}) p(\mathbf{v}) d\mathbf{x} d\mathbf{v} \quad (26)$$

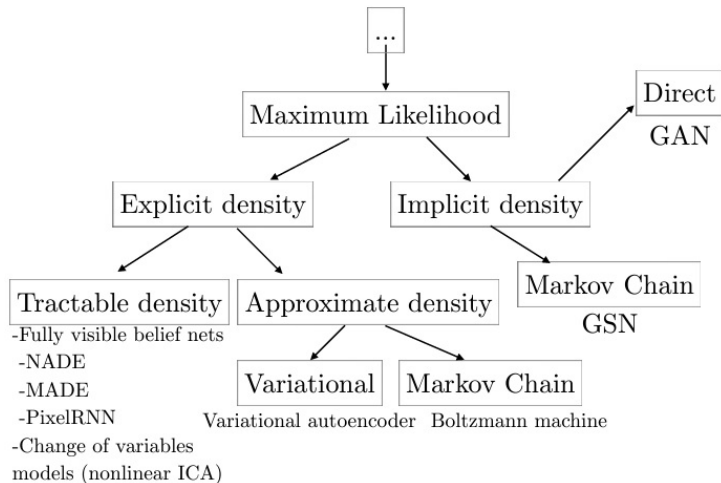
up to a constant addition and sign-preserving multiplication.

- ▶ Under certain conditions, the minimizer of SFD is consistent and holds asymptotic normality.

Concluding Remark



Concluding Remark



The figure is from Goodfellow (2016).

Concluding Remark



- ▶ We have reviewed recent developments in deep generative models, with a particular focus on targeted statistical distances.
- ▶ There are many ways to explore recent advancements in deep generative models. Examples include developing model classes for specific data types, analyzing the properties of model classes and estimators, or adapting large pre-trained models to new datasets.

Concluding Remark

- ▶ Such changes may not always be exciting for everyone.
- ▶ Rapid advancements inevitably bring anxiety and uncertainty as the expectations of academia and industry continue to evolve.⁵
- ▶ I hope that this course has helped you find joy and beauty in deep generative models and has also encouraged you to continue exploring new waves in this evolving field.

⁵ *"i sensed anxiety and frustration at NeurIPS'24"*

<https://kyunghyuncho.me/i-sensed-anxiety-and-frustration-at-neurips24/>

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